SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: ITA 06 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 10%

QUESTIONS: 15 Total Marks: 25 Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Nirmal kumar / 192412003 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

GPS Photo


SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.
Evolutionary Machine Learning Platform That Automaticall
Discovers the Best Model for Any Dataset
Presented by
Name: Nirmal Kumar T
Register No: 192412003
Dr K Ba
Department of Comp
SIMA
GPS Map Camera
Chennai,Tamil Nadu,India
Thandalam Sriperumbadur, Chennai, Tamil Nadu 602105, India
Lat 13.026475, Long 80.016290
Tuesday, 28/07/2026 08:38 AM GMT+05:30
Note : Captured by GPS Map Camera

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.

Evolutionary Machine Learning Platform That Automatically Discovers the Best Model for Any Dataset

Presented by

Name: Nirmal Kumar T
Register No: 192412003

Supervisor
Dr K Baskar, Professor
Department of Computer Science and Engineering
SIMATS Engineering

Abstract

- Developed an intelligent AutoML platform that automatically analyzes datasets, performs preprocessing, and identifies the most suitable machine learning algorithm.
- Uses evolutionary optimization techniques to compare multiple models and optimize hyperparameters for improved prediction accuracy.
- Supports classification and regression problems with minimal human intervention, making machine learning accessible to non-experts.
- Generates performance metrics, model comparison reports, and recommends the best-performing model.

- The platform reduces model selection time, improves efficiency, and contributes to SDG 9 (Industry, Innovation and

Problem Statement

- Selecting the best machine learning model requires significant expertise, time, and repeated experimentation.
- Manual preprocessing, feature selection, model selection, and hyperparameter tuning are complex and error-prone.
- Different datasets require different algorithms, making it difficult for beginners and organizations to identify the optimal model.
- Existing approaches are often time-consuming and may not achieve the best prediction accuracy.

Objectives

- To develop an Evolutionary Machine Learning Platform that automatically performs data preprocessing, identifies the problem type, trains multiple machine learning models, and recommends the best-performing model based on evaluation metrics.
- To improve prediction accuracy and reduce model development time by optimizing machine learning algorithms using evolutionary techniques and evaluating their performance with metrics such as Accuracy, Precision, Recall, F1-Score, RMSE, and R^2 Score.

Literature Survey

Sl. No	Year of Publication	Authors	Title of the Publication	Findings	Limitations
1	2024	Mitra Baratchi et al.	Automated Machine Learning: Past, Present and Future	Provides a comprehensive overview of AutoML techniques, including automated model selection and hyperparameter optimization.	Mainly a survey paper; does not propose a new AutoML framework. (Springer)
2	2024	Rafael Barbudo et al.	Evolving Machine Learning Workflows Through Interactive AutoML	Uses evolutionary algorithms to optimize ML workflows with human interaction, reducing search time while maintaining accuracy.	Requires user interaction and is not fully automated. (arXiv)
3	2024	Nikolay O. Nikitin et al.	Integration of Evolutionary Automated Machine Learning with Structural Sensitivity Analysis for Composite Pipelines	Improves AutoML pipeline optimization using evolutionary search and sensitivity analysis.	Computational cost increases for complex and large-scale datasets. (ScienceDirect)
4	2025	Keshav Das et al.	AutoML Algorithms for Online Generalized Additive Model Selection	Applies evolutionary AutoML to automatically select high-performing forecasting models.	Focused on electricity demand forecasting rather than general-purpose datasets. (Proceedings of Machine Learning Research)
5	2025	Patara Trirat et al.	AutoML-Agent: A Multi-Agent LLM Framework for Full-Pipeline AutoML	Introduces multiple AI agents to automate preprocessing, model selection, and deployment.	High computational requirements due to multiple LLM agents. (Proceedings of Machine Learning Research)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**35**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**25**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**10**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand